Sequential Time-Warping Hidden Markov Model

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September 2026

1 Introduction

Learning and memory support flexible animal behavior by enabling animals to use past experiences to predict the consequences of future actions (O'Keefe & Nadel, 1978). During consummatory behaviors and NREM sleep, theta sequences are replayed during sharp wave-ripples (SPW-Rs) and are thought to reflect a memory consolidation process (Wilson & McNaughton, 1994). However, capturing the statistical dynamics of theta sequences has been primarily limited to sequences of place cells, which only account for a minority of SPW-Rs (Buzsaki, 2015). Furthermore, the rate of sequential place cell progression is faster during SPW-Rs compared to theta cycles (Diba & Buzsaki, 2007) and other factors, such as brain temperate, can effect the frequency of ongoing rhythms (Peterson & Buzsaki, 2020; Peterson, Voroslakos & Buzsaki 2022), putatively modulating the rate of sequence progression. To address these challenges, we introduce an unsupervised learning algorithm which captures sequential strucure in neuronal spike trains and explicitly models time-warping.

2 Method

2.1 Observations and Emission Model

Suppose we observe spike counts Y across a population of N neurons for T time steps, such that

$$Y = (\vec{y}_1, \vec{y}_2, \ldots, \vec{y}_T)\,, \tag{1}$$

where each observation $\vec{y}_t$ is a vector of spike counts across the N neurons,

$$\vec{y}_t = \begin{bmatrix} y_t^1 \\ y_t^2 \\ \vdots \\ y_t^N \end{bmatrix} \in \mathbb{N}^N. \tag{2}$$

We assume that the dynamics of a set of M latent states $Q = \{1, 2, \ldots, M\}$ generate the spike counts which we assume are conditionally independent and drawn from a Poisson distribution. That is, given the latent state sequence

$$Z = z_1, z_2, ..., z_T \tag{3}$$

we have that

$$b_i(\vec{y}_t) := p(\vec{y}_t \mid z_t = i) = \prod_{n=1}^{N} \frac{(\lambda_{ni})^{y_t^n} e^{-\lambda_{ni}}}{y_t^n!} \tag{4}$$

and the rates $\{\lambda_{ni}\}$ are stored in the emissions matrix $\Lambda \in \mathbb{R}_+^{N \times M}$ such that,

$$\Lambda_{ni} = \lambda_{ni} \tag{5}$$

2.2 Sequential Latent-State Dynamics

The dynamics of the latent states are determined probabilistically, but are assumed to be constrained to ordered sequences such that the only allowed transitions are $q_1 \to q_j$, $q_i \to q_1$, and $q_i \to q_{i+1}$. This can be achieved if we constrain

$$p(z_{t+1} = q_{i+k} \mid z_t = q_i) = 0 \tag{6}$$

except when $k = 1$, $i = 1$ or when $i + k = 1$ which gives us a matrix $T^\star$ with the following form

$$T^\star = \begin{bmatrix} 0 & T_{12} & T_{13} & T_{14} & \cdots & T_{1M} \\ T_{21} & 0 & 0 & 0 & \cdots & 0 \\ T_{31} & T_{32} & 0 & 0 & \ddots & \vdots \\ T_{41} & 0 & T_{43} & 0 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & 0 \\ T_{M1} & 0 & 0 & 0 & T_{M,M-1} & 0 \end{bmatrix} \in \mathbb{R}^{M \times M}. \tag{7}$$

where,

$$T_{ij}^\star = p(z_{t+1} = q_i \mid z_t = q_j) \tag{8}$$

2.3 Latent Time-Warping Dynamics

Now, in order to explicity model dynamical time-warping of sequences, we introduce a second latent variable $\tau \in \mathbb{S} = \{s_1, s_2, ..., s_S\}$ where $s_k \in [0, 1]$ which evolves according to its own transition matrix H

$$H_{ij} = p(\tau_{t+1} = s_i \mid \tau_t = s_j) \tag{9}$$

and warps time by changing the probability of self transitions through the following equation,

$$T_{ij}^{(\omega)} = (1 - s_\omega) I_{ij} + s_\omega T_{ij}^\star \tag{10}$$

Intuitively, the time-warp τ slows down sequence transitions by changing the

2.4 Joint Latent-State Transition Model

From here we have introduced all the latent variables which we hold in the Master Tensor $\mathbb{P} \in \mathbb{R}^{M \times S \times M \times S}$ which tells is the the probabilities of all possible combinations of latent state transitions,

$$\mathbb{P}_{ijl\omega} = p((z_{t+1} = q_i) \cap (\tau_{t+1} = s_j) \mid (z_t = q_l) \cap (\tau_t = s_\omega)) \tag{11}$$

Now since τ evolves independently of z while z depends both on itself and τ, we can write,

$$\mathbb{P}_{ijl\omega} = p(\tau_{t+1} = s_j \mid \tau_t = s_\omega) p(z_{t+1} = q_i \mid z_t = q_l \cap \tau_t = s_\omega) \tag{12}$$

By equations (9) and (10) we have that

$$\begin{aligned} \mathbb{P}_{ijl\omega} &= H_{j\omega} T_{il}^{(\omega)} && (13) \\ &= H_{j\omega}[(1 - s_\omega) I_{il} + s_\omega T_{il}^{\star}] && (14) \end{aligned}$$

2.5 Generative Dynamics

To show how the generative model updates we will introduce a joint probability distribution $(P_t) \in \mathbb{R}^{M \times S}$ which tells us

$$(P_t)_{ij} = p((z_t = q_i) \cap (\tau_t = s_j)) \tag{15}$$

To show how (P_t) updates we just need to use the law of total probability,

$$(P_{t+1})_{ij} = \sum_{l=1}^{M} \sum_{\omega=1}^{S} \mathbb{P}_{ijl\omega} (P_t)_{l\omega} \tag{16}$$

Using (14) we have that

$$(P_{t+1})_{ij} = \sum_{l=1}^{M} \sum_{\omega=1}^{S} H_{j\omega}[(1 - s_\omega) I_{il} + s_\omega T_{il}^{\star}] (P_t)_{l\omega} \tag{17}$$

2.6 Complete-Data Likelihood

With all of this in place we can derive the full data-likelihood. First we will define all the parameters of the model as $\Theta = [\Lambda, T^{\star}, H, \pi]$ where $\pi \in \mathbb{R}^{M \times S}$ is the initial probability distribution,

$$\pi_{ij} = p(z_1 = i, \tau_1 = j) \tag{18}$$

Then we just need to loop through all the time points multiplying the probabilities of latent state transitions with the probabilities of observations given the latent states,

$$p(Y, Z, \tau \mid \Theta) = p(z_1, \tau_1) \prod_{t=1}^{T-1} p(z_{t+1}, \tau_{t+1} \mid z_t, \tau_t) \prod_{t=1}^{T} b_{z_t}(\vec{y}_t) \tag{19}$$

$$= \pi_{z_1 \tau_1} \prod_{t=1}^{T-1} \mathbb{P}_{z_{t+1} \tau_{t+1} z_t \tau_t} \prod_{t=1}^{T} b_{z_t}(\vec{y}_t) \tag{20}$$

$$= \pi_{z_1 \tau_1} \prod_{t=1}^{T-1} H_{\tau_{t+1} \tau_t}[(1 - s_{\tau_t}) I_{z_{t+1}, z_t} + s_{\tau_t} T^{\star}_{z_{t+1}, t}] \prod_{t=1}^{T} b_{z_t}(\vec{y}_t) \tag{21}$$

2.7 Maximum Posterior Decoding

To infer the latent states Z and τ given an observation sequence Y and the model parameters Θ, we will find the maximum a posterior estimate of the latent states given the observations. To do this, lets define $v_t(l, \omega)$ as the probability of being in the latent states $\{z_t = l, \tau_t = \omega\}$ after passing through the most likely sequence of latent states and observing the spike counts $y_1, \ldots, y_t$. That is,

$$v_t(i, j) = \max_{\{z_k, \tau_k\}_{k=1,\ldots,t-1}} p(\{z_k, \tau_k, \vec{y_k}\}_{k=1,\ldots,t-1}, \vec{y_t}, z_t = i, \tau_t = s_j \mid \Theta) \tag{22}$$

which we can solve inductively,

$$v_1(i, j) = \pi_{ij} b_i(\vec{y}_1) \tag{23}$$

$$v_{t+1}(i, j) = \max_{\{l\}_{l=1,\ldots,M}\{\omega\}_{1,\ldots,S}} [v_t(l, \omega) \mathbb{P}_{ijl\omega} b_i(\vec{y}_{t+1)}] \tag{24}$$

And we just need to get the back-trace pointers so that we can trace the most likely sequence,

$$\psi_{t+1}(i, j) = \arg\max_{\{l\}_{l=1,\ldots,M}\{\omega\}_{1,\ldots,S}} [v_t(l, \omega) \mathbb{P}_{ijl\omega}] \tag{25}$$

Then, we just need to initialize the back-trace with,

$$\{\hat{z}_T, \hat{\tau}_T\} = \arg\max_{\{l\}_{l=1,\ldots,M}\{\omega\}_{1,\ldots,S}} v_T(i, j) \tag{26}$$

and we can trace backwards the most likely hidden state sequence,

$$\{\hat{z}_{t-1}, \hat{\tau}_{t-1}\} = \psi_t(\hat{z}_t, \hat{\tau}_t) \tag{27}$$

Thus, we have shown how to find the most likely latent state sequence given the model parameters and the observations.

2.8 Parameter Estimation with Expectation-Maximization

We will now show how to learn the parameters of the model. That is, we will estimate transition probabilities $T^\star$ the emissions Λ and the time-warping matrix H. To do this lets define the model parameters as $\Theta = [T^\star, \Lambda, H]$. We will first need to define the forward and backward probabilties which will be used for computing the expected emissions and transitions probabilities.

First we will define the forward probability, which is the probability of being in a particular state pair $(z_t = i, \tau_t = j)$ at time t and observing all the spike counts up to time t, given the model parameters. That is,

$$\alpha_t(i,j) := p(\vec{y}_1, \ldots, \vec{y}_t, z_t = i, \tau_t = j \mid \Theta) \tag{28}$$

which can be solved inductively,

$$\alpha_1(i,j) = \pi_{ij} b_i(\vec{y}_1) \tag{29}$$

$$\alpha_{t+1}(i,j) = b_i(\vec{y}_{t+1}) \sum_{l=1}^{M} \sum_{\omega=1}^{S} \alpha_t(l,\omega) \mathbb{P}_{ijl\omega} \tag{30}$$

Next we define the backward probability, which is the probability of being in a particular state pair $(z_t = i, \tau_t = j)$ at time t and observing all the spike counts after time t, given the model parameters Θ. That is,

$$\beta_t(i,j) := p(\vec{y}_{t+1}, \ldots, \vec{y}_T \mid z_t = i, \tau_t = j, \Theta) \tag{31}$$

which can also be solved inductively,

$$\beta_T(i,j) = 1 \tag{32}$$

$$\beta_t(l,\omega) = \sum_{i=1}^{M} \sum_{j=1}^{S} \beta_{t+1}(i,j) \mathbb{P}_{ijl\omega} b_i(\vec{y}_{t+1}) \tag{33}$$

Now we can use the forward and backward probabilities for the expectation-step before updating the model parameters to maximize this value. To estimate the expected number of transitions between state pairs we will define $\zeta_t(i,j,l,\omega)$ as the probability of transitioning from the state pair $(z_t = l, \tau_t = s_\omega) \to (z_{t+1} = i, \tau_{t+1} = s_j)$ given the observations Y and the model parameters Θ. Thus, we can write,

$$\zeta_t(i,j,l,\omega) := p(z_{t+1} = i, \tau_{t+1} = s_j, z_t = l, \tau_t = s_\omega \mid Y, \Theta) \tag{34}$$

$$= \frac{p(z_{t+1} = i, \tau_{t+1} = s_j, z_t = l, \tau_t = s_\omega, Y \mid \Theta)}{(Y \mid \Theta)} \tag{35}$$

$$= \frac{\alpha_t(l,\omega) \beta_{t+1}(i,j) \mathbb{P}_{ijl\omega} b_i(\vec{y}_{t+1})}{\sum_{k=1}^{M} \sum_{r=1}^{S} \alpha_t(k,r) \beta_t(k,r)} \tag{36}$$

First, we can update the time-warp matrix by defining $\varphi_t(i,j)$ as the probability of transitioning from state $\tau_t = j \to \tau_{t+1} = i$ given all the observations

Y and the model parameters Θ. This can be achieved simply by marginalizing ζ_t over the latent states z_t and z_{t+1},

$$\varphi_t(i,j) := p(\tau_{t+1} = s_i, \tau_t = s_j \mid Y, \Theta) \tag{37}$$

$$= \sum_{l=1}^{M} \sum_{l'=1}^{M} \zeta_t(l, i, l', j) \tag{38}$$

And we can update the time-warp matrix with,

$$\hat{H}_{ij} = \frac{\sum_{t=1}^{T-1} \varphi_t(i,j)}{\sum_{t=1}^{T-1} \sum_{k=1}^{S} \varphi_t(k,j)} \tag{39}$$

Next, we will work on updating the transition matrix $T^\star$ which can be done similarly to the time-warp matrix H. Since we have constrained $T^\star_{ii} = 0$, we only need to learn the non-diagonal transition probabilities

$$T^\star_{ij} = \frac{T^{(\omega)}_{ij}}{s_\omega} : i \neq j \tag{40}$$

Suppose a particular transition takes place,

$$z_t = j \rightarrow z_{t+1} = i : i \neq j \tag{41}$$

and,

$$\tau_t = s_\omega \tag{42}$$

then the contribution of this particular transition to the log likelihood becomes

$$\log \hat{T}^{(\omega)}_{ij} = \log \hat{T}^\star_{ij} + \log s_\omega \tag{43}$$

where $\log s_\omega$ does not depend on $T^\star_{ij}$ and therefore does not contribute to the derivative of the expected complete-data log likelihood with respect to $T^\star_{ij}$. With this knowledge we can marginalize over the time-warps with

$$\xi_{ij} = \sum_{t=1}^{T-1} \sum_{\omega=1}^{S} \sum_{k=1}^{S} \zeta_t(i, k, j, \omega) \tag{44}$$

and update the transition matrix accordingly,

$$\hat{T}^\star_{ij} = \frac{\xi_{ij}}{\sum_{l=1}^{M} \xi_{lj}} \tag{45}$$

In theory, the zero elements of $T^\star$ will remain zero for each step since these expected transitions remain zero. However, in practice, especially while working in log-space, these values will need to be clamped to zero with a mask on each update.

Finally, to update the emissions matrix Λ we just need to find the expected rate in each latent state. To do this, lets define $\gamma_t(i)$ as the probability of being in state $z_t = i$ given the observation sequence Y and the model parameters Θ.

$$\gamma_t(i) := p(z_t = i \mid Y, \Theta) \tag{46}$$

$$= p(z_t = i, Y \mid \Theta) p(Y \mid \Theta) \tag{47}$$

$$= \frac{\sum_{\omega=1}^{S} \alpha_t(i, \omega) \beta_t(i, \omega)}{\sum_{\omega'=1}^{S} \sum_{k=1}^{M} \alpha_t(k, \omega') \beta_t(k, \omega')} \tag{48}$$

And the emissions matrix can be updated with a weighted mean over the spike counts Y,

$$\hat{\Lambda}_{nj} = \frac{\sum_{t=1}^{T} y_t^n \gamma_t(j)}{\sum_{t=1}^{T} \gamma_t(j} \tag{49}$$

Thus, we have shown how to update all the model parameters with Expectation-Maximization.